Sardar Vallabhbhai National Institute of Technology
Surat-395007, Gujarat, India
Department of Electronics Engineering

EC343 : Antenna Theory

GROUP-1
PROJECT

Entitled

Design and simulation of a dual helix helical antenna in CST

Project Members

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Dual Helix Antenna Design

Acknowledgment

We would like to express our sincere gratitude to our lecturer, **Dr. Suresh Dahiya**, and **Aman Varshney** for their valuable guidance, continuous encouragement, and support throughout the course of this project. Their insights were instrumental in the successful design and simulation of the Dual Helix Antenna.

Abstract

This report details the design, theoretical analysis, and simulation of a Dual Helix Antenna operating in the L1 GPS frequency band (1.57542 GHz). The study focuses on achieving Circular Polarization (CP), specifically Right-Hand Circular Polarization (RHCP), to mitigate polarization mismatch loss and Faraday rotation effects common in satellite communications. We utilize CST Studio Suite to simulate various configurations, including single and dual helix structures with different wavelength parameters (λ and 2λ). The report presents the derivation of key physical parameters—such as helix radius, pitch angle, and number of turns—and includes transmission line calculations required for impedance matching and phase shifting.

Contents

1 Introduction **3**

2 Theory **3**
- 2.1 Dual Helix Antenna . . . 3
- 2.2 Axial Mode Radiation . . . 3
- 2.3 Transmission Line using Meander Line . . . 4

3 Design Methodology **4**

4 Design Parameters and Calculations **4**
- 4.1 Wavelength Calculation . . . 4
- 4.2 Design Equations . . . 4
- 4.3 Calculated Configurations . . . 5

5 Transmission Line Calculations **5**
- 5.1 Free-space Wavelength λ_0 . . . 5
- 5.2 Effective Dielectric Constant ε_{eff} . . . 6
- 5.3 Guided Wavelength λ_g . . . 6
- 5.4 Axial Length for Phase Advance . . . 6

6 Design Steps **7**

7 Simulation Results **8**
- 7.1 Single Helical Antenna . . . 8
- 7.2 Dual Helical Antenna (Configuration λ) . . . 9
- 7.3 Dual Helical Antenna (Configuration 2λ) . . . 10
- 7.4 Dual Helical Antenna with Transmission Line (λ) . . . 11
- 7.5 Parametric Study: Height Variation . . . 12
- 7.6 Parametric Study: Ground Plane Variation . . . 13

8 Conclusion **14**

9 References **15**

1 Introduction

The primary objective of this project is the design and simulation of a dual helix helical antenna. The design is optimized for the L1 GPS band, widely used by GNSS systems including GPS, Galileo, and QZSS for navigation and timing.

CST Studio Suite (CST Microwave Studio) was utilized as the primary simulation tool to model the antenna structure and analyze its electromagnetic properties, including S-parameters, radiation patterns, and field distributions.

2 Theory

2.1 Dual Helix Antenna

A **Dual Helical Antenna** consists of two helical conductors wound in parallel around a common axis. These helices may be fed in-phase or with a specific phase shift to enhance circular polarization performance. Similar to a single helix, the dual configuration can operate in *axial mode* for end-fire circularly polarized radiation or in *normal mode* for broadside radiation. The dual-helix structure improves performance by providing wider bandwidth, more stable input impedance, reduced cross-polarization, and better polarization purity compared to a single helix.

Key Parameters:

- **Helix radius (R):** Affects beamwidth and impedance.
- **Pitch angle (α):** Determines axial mode radiation efficiency.
- **Number of turns (N):** Controls gain and directivity.
- **Spacing between helices (S):** Influences coupling and polarization purity.

Operation in L1 Band: In this work, the dual helical antenna is designed to operate in the **L1 GPS band** at **1.57542 GHz**. Since GNSS satellites transmit **Right-Hand Circularly Polarized (RHCP)** signals, the dual helix—operating in axial mode—naturally supports RHCP, making it highly suitable for reliable satellite signal reception.

2.2 Axial Mode Radiation

The operational mode of a helical antenna is determined by the relationship between its physical dimensions and the operating wavelength. When designed to operate in the **axial mode**, the helix functions as an efficient end-fire radiator.

The adoption of circular polarization is critical to mitigate:

1. **Polarization Mismatch Loss:** Misalignment caused by a tumbling satellite or mobile user orientation can lead to signal fade. CP eliminates this dependency.
2. **Faraday Rotation Mitigation:** The ionosphere twists the polarization plane of linear waves. CP waves are inherently immune to this degradation.

2.3 Transmission Line using Meander Line

A **meander line** is a compact folded wire antenna or transmission line where the conductor is bent back and forth to reduce physical length while maintaining electrical length.

Key Parameters:

- **Effective length:** $L_{\text{eff}} \approx \frac{\lambda}{2}$ (for resonance).
- **Resonant frequency:** $f_r = \frac{c}{2L_{\text{eff}}\sqrt{\epsilon_{\text{eff}}}}$.
- **Impedance:** $Z \approx 50 + jX\ \Omega$, depends on line spacing and width.

3 Design Methodology

The design process in CST Studio Suite involved the following key steps to create the dual helix structure:

1. **Ground and Substrate:** Defining the base ground plane and dielectric substrate.
2. **Helix Creation:** Using the CST Macro tool to generate the helical coil structure.
3. **Dual Configuration:** Mirroring the generated spiral by 180° in the Z-plane to create the bifilar structure.
4. **Port Creation:** Defining the excitation ports for the antenna feeds.
5. **Transmission Line:** Creating the transmission line network for impedance matching and phase shifting.
6. **Discrete Port:** Picking the relevant faces and applying discrete ports for simulation.

4 Design Parameters and Calculations

4.1 Wavelength Calculation

$$
\begin{aligned}
f &= 1.57542 \times 10^9 \text{ Hz (GPS L1 frequency)} \\
c &= 3 \times 10^8 \text{ m/s} \\
\lambda &= \frac{c}{f} = \frac{3 \times 10^8}{1.57542 \times 10^9} \approx 0.19057 \text{ m} = 190.57 \text{ mm}
\end{aligned}
$$

4.2 Design Equations

$$
\begin{aligned}
C &= \text{Circumference of helix} \\
R &= \frac{C}{2\pi} \quad \text{(Helix radius)} \\
S &= C \cdot \tan\alpha \quad \text{(Pitch spacing between turns)} \\
H &= N \cdot S \quad \text{(Height of the helix)}
\end{aligned}
$$

4.3 Calculated Configurations

Case 1: Helix with Circumference λ

$$
\begin{aligned}
C &= \lambda = 190.57 \text{ mm} \\
R &= \frac{190.57}{2\pi} \approx 30.34 \text{ mm} \\
S &= 190.57 \cdot \tan 12^\circ \approx 40.42 \text{ mm} \\
H &= 6 \cdot 40.42 \approx 242.52 \text{ mm}
\end{aligned}
$$

Case 2: Helix with Circumference 2λ

$$
\begin{aligned}
C &= 2\lambda = 381.15 \text{ mm} \\
R &= \frac{381.15}{2\pi} \approx 60.63 \text{ mm} \\
S &= 381.15 \cdot \tan 12^\circ \approx 80.85 \text{ mm} \\
H &= 6 \cdot 80.85 \approx 485.10 \text{ mm}
\end{aligned}
$$

Case 3: Helix with Fixed Radius $r = 50$ mm

$$
\begin{aligned}
C &= 2\pi(50) \approx 314.16 \text{ mm} \\
S &= 314.16 \cdot \tan 15^\circ \approx 84.20 \text{ mm} \\
H &= 6 \cdot 84.20 \approx 505.21 \text{ mm}
\end{aligned}
$$

Table 1: Summary of Design Parameters

Parameter	**For λ**	**For 2λ**	**For $r = 50$ mm**
Circumference C (mm)	190.57	381.15	314.16
Radius R (mm)	30.34	60.63	50.00
Pitch spacing S (mm)	40.42	80.85	84.20
Number of turns N	6	6	6
Height H (mm)	242.52	485.10	505.21
Wire radius r_{wire} (mm)	0.5	0.5	0.5
Pitch angle α (deg)	12	12	15

5 Transmission Line Calculations

5.1 Free-space Wavelength λ_0

$$
\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8}{1.57542 \times 10^9} = 190.43 \text{ mm}
$$

5.2 Effective Dielectric Constant ε_{eff}

Given $\varepsilon_r = 4.4$ (FR4), $h = 1.6$ mm, $W = 3.06$ mm:

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \left[\frac{1}{\sqrt{1 + 12\frac{h}{W}}} \right] \approx 3.33$$

5.3 Guided Wavelength λ_g

$$\lambda_g = \frac{\lambda_0}{\sqrt{\varepsilon_{\text{eff}}}} = \frac{190.43}{\sqrt{3.33}} = 104.35 \text{ mm}$$

Quarter-wave and half-wave calculations:

$$\frac{\lambda_g}{4} = 26.09 \text{ mm}, \qquad \frac{\lambda_g}{2} = 52.17 \text{ mm}$$

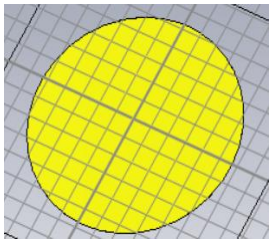
5.4 Axial Length for Phase Advance

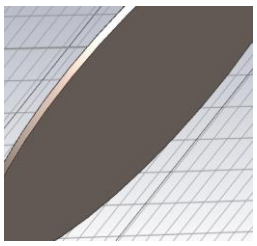
For an axial-mode helical antenna, the phase advance along the axis per turn is $\beta \cdot l = \pi$.

$$l = \frac{\pi\lambda}{2\pi} = \frac{\lambda}{2}$$

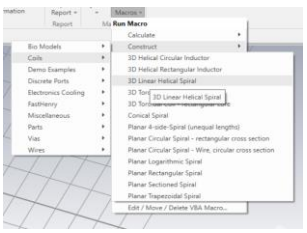
This confirms the axial length per turn required for a 180° phase shift.

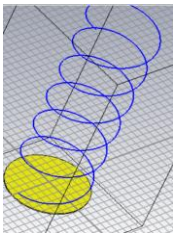
Design Steps

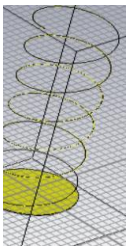




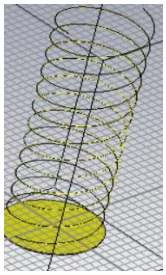
1) Ground and Substrate



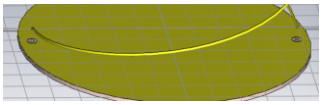


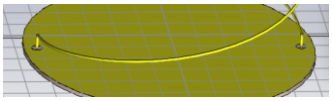


2) Using Macro to create Helix

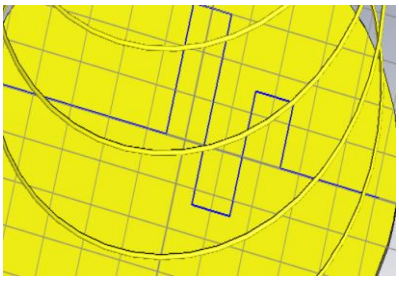


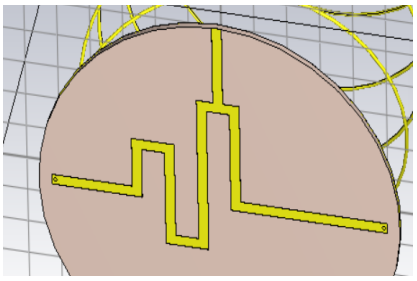
3) Mirror the Spiral by 180 in z-plane



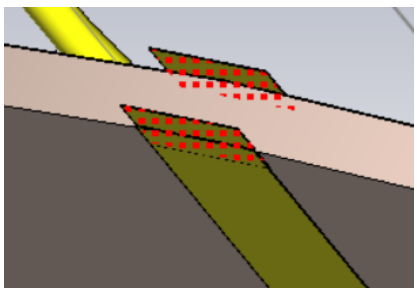


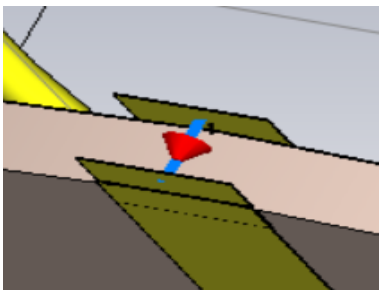
4) Creating the ports



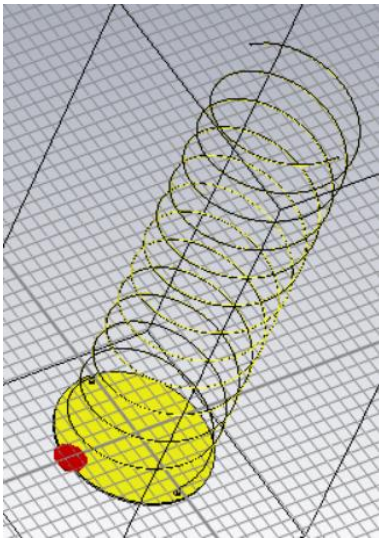


5) Creating Transmission line





6) Picking Face and applying discrete port

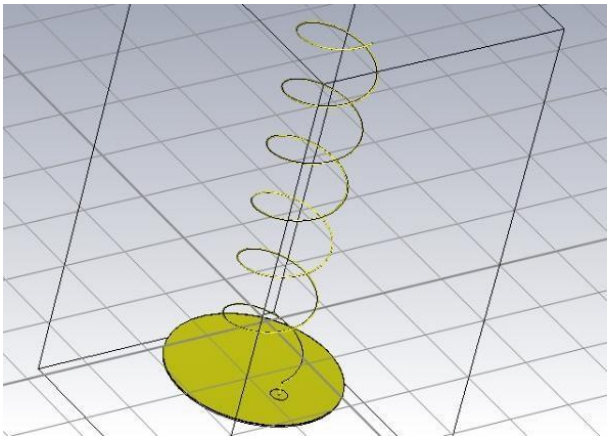


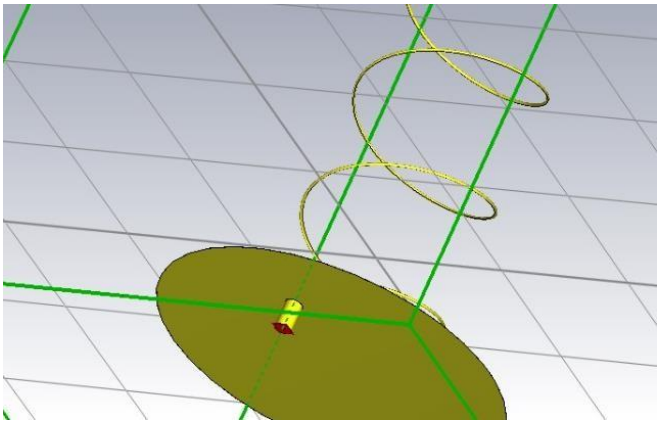
7) Final model

Antenna Results and Simulation Outputs:

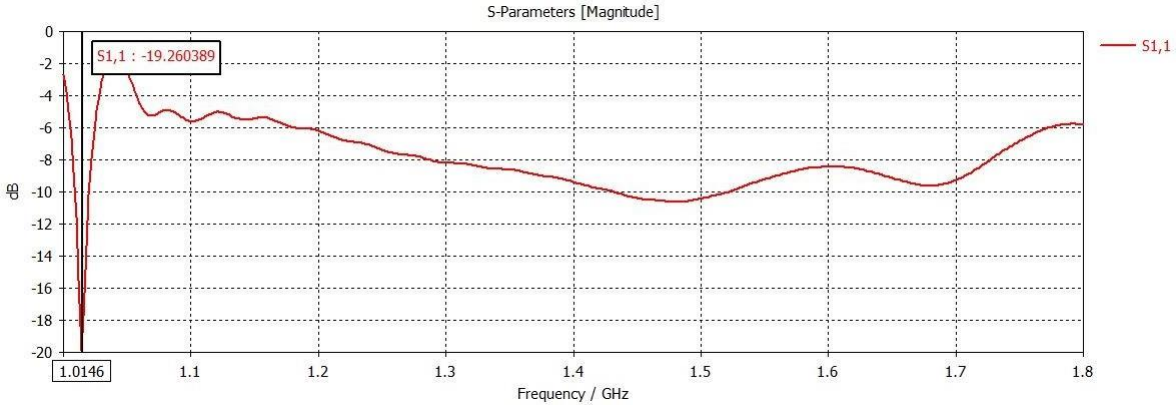
1. Single Helical Antenna:

3D Model:

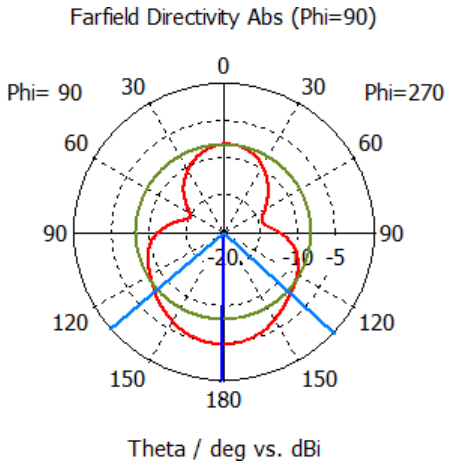




S-Parameter:



Farfield Directivity:

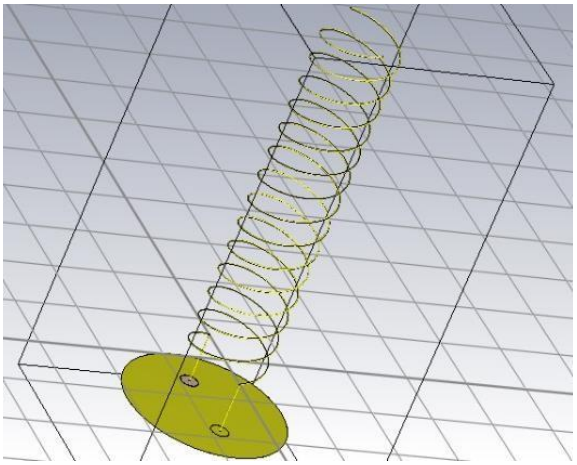


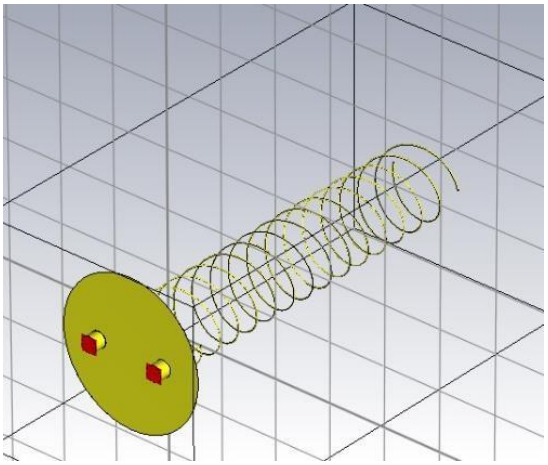


Frequency = 1 GHz
Main lobe magnitude = -4.96 dBi
Main lobe direction = 179.0 deg.
Angular width (3 dB) = 96.6 deg.
Side lobe level = -3.2 dB

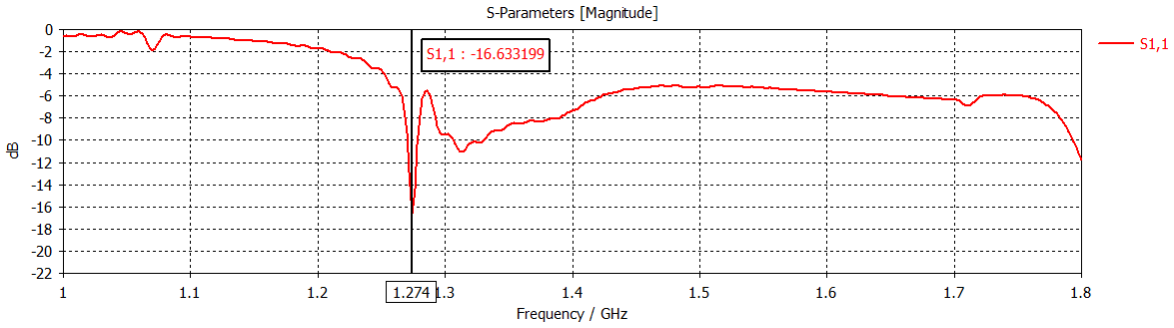
2. Dual Helical Antenna (without transmission line for λ):

3D Model:



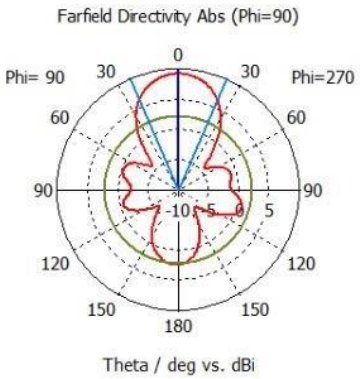


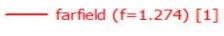
S-Parameter:



7

Farfield Directivity:

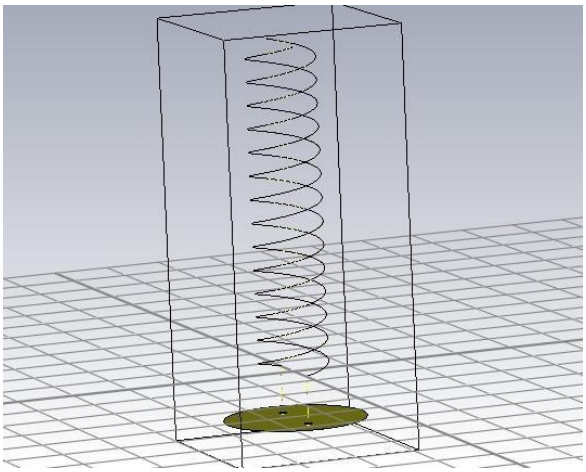


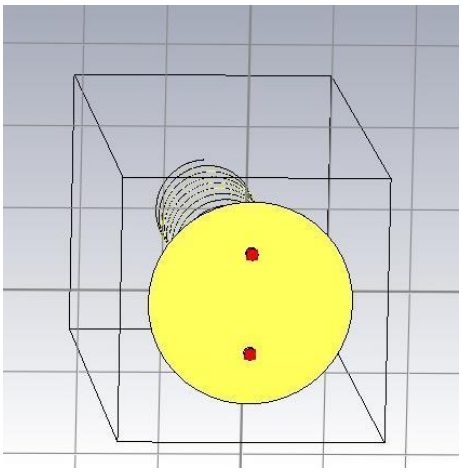


Frequency = 1.274 GHz
Main lobe magnitude = 7.28 dBi
Main lobe direction = 4.0 deg.
Angular width (3 dB) = 55.5 deg.
Side lobe level = -4.0 dB

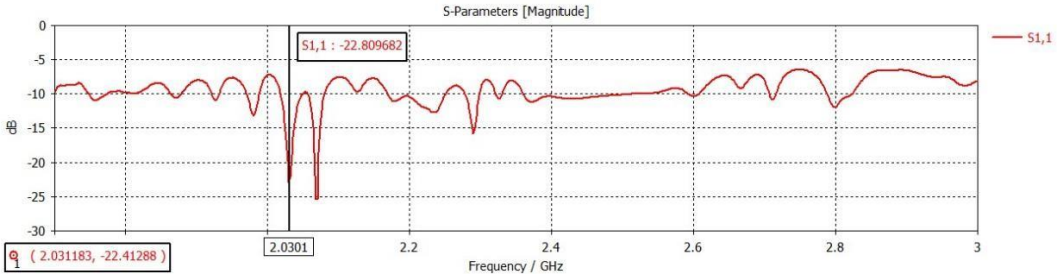
3. Dual Helical Antenna (without transmission line for 2λ):

3D Model:

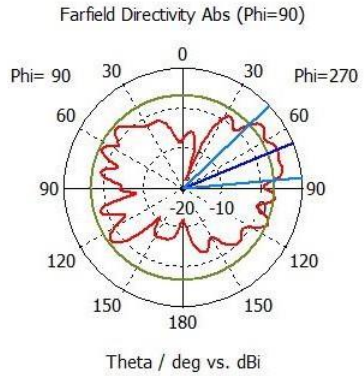


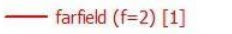


S-Parameter:



Farfield Directivity:

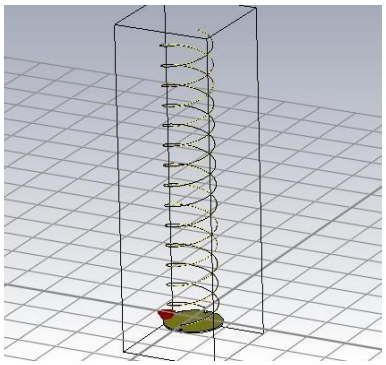


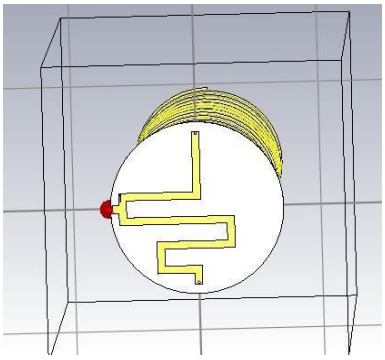


Frequency = 2 GHz
Main lobe magnitude = 5.7 dBi
Main lobe direction = 68.0 deg.
Angular width (3 dB) = 38.8 deg.
Side lobe level = -2.5 dB

4. Dual Helical Antenna (with transmission line for λ):

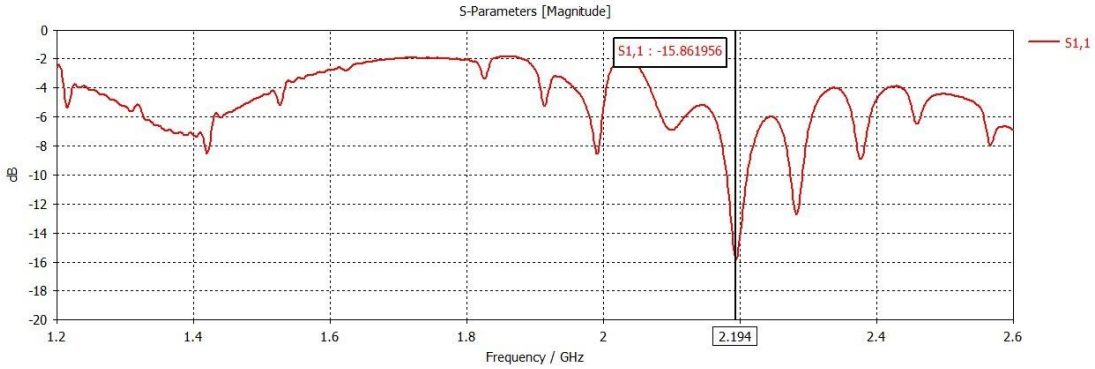
3D Model:



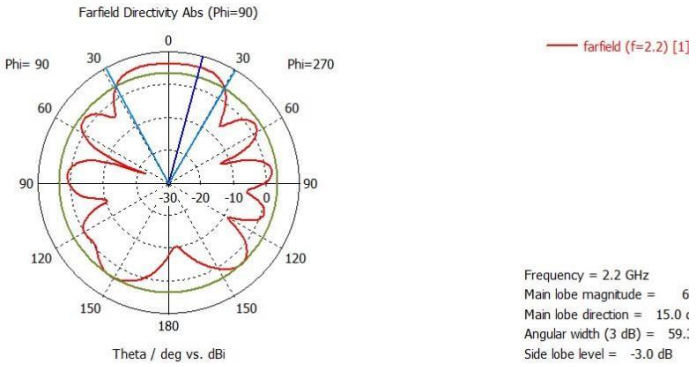


Parameters:
h = 242.52mm
N = 6
α = 15
s = 40.42mm
R = 50mm
R-wire = 0.5mm
R-gnd = 55mm

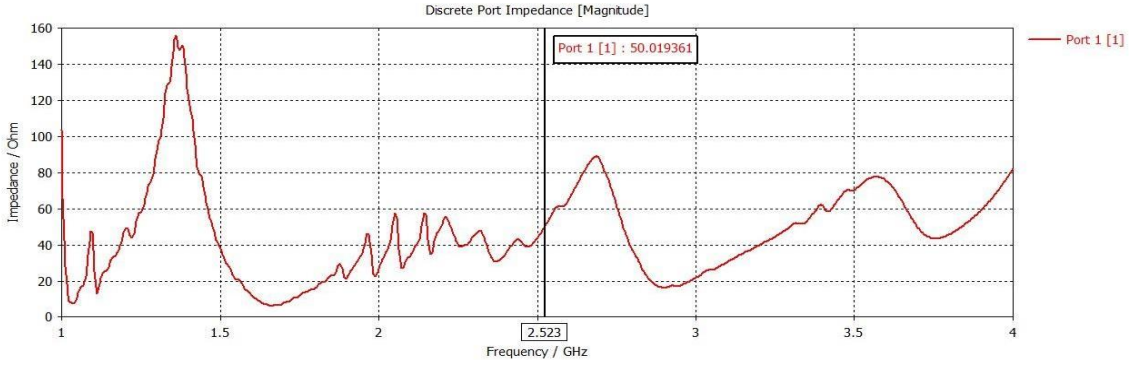
S-Parameter:



Farfield Directivity:

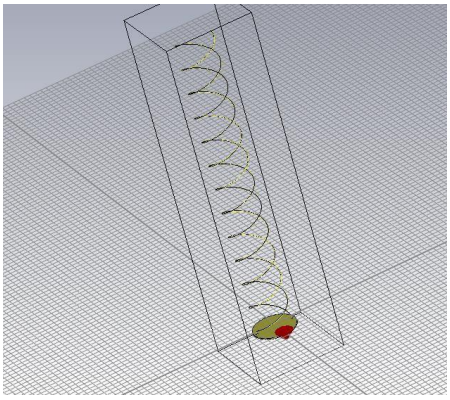


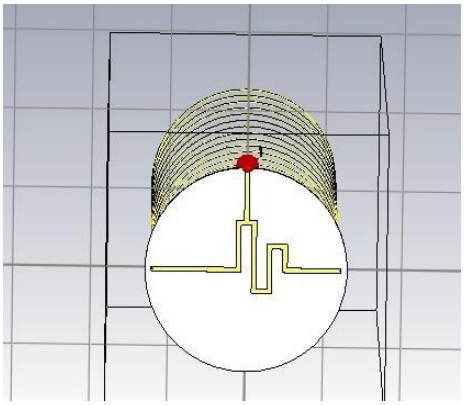
Impedance:



Changing Height to 500mm:

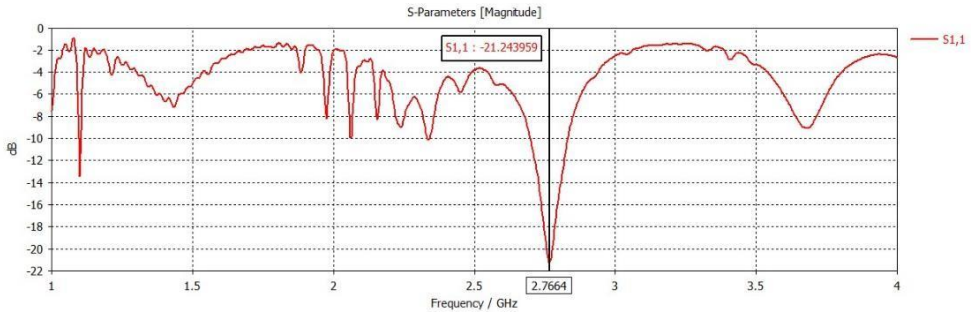
3D Model:



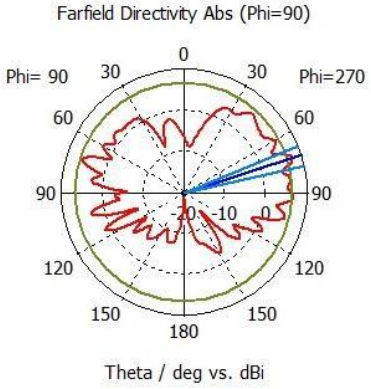


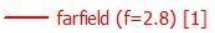
Parameters:
h = 500mm
N = 6
α = 15
s = 40.42mm
R = 30.33mm
R-wire = 0.5mm
R-gnd = 55mm

S-Parameter:



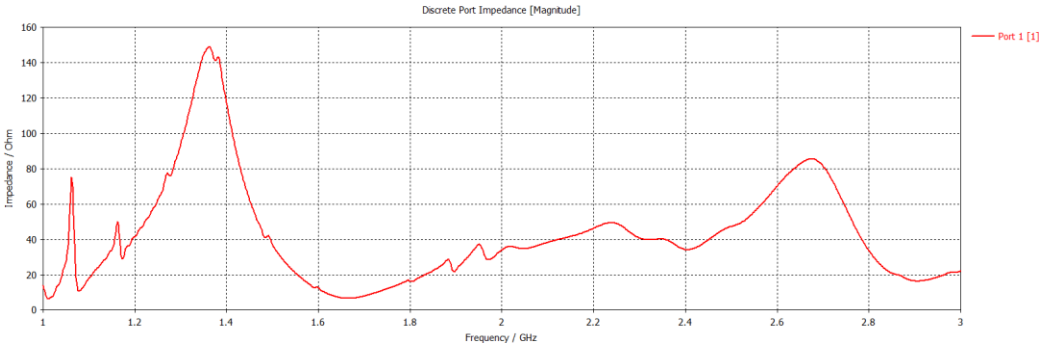
Farfield Directivity:





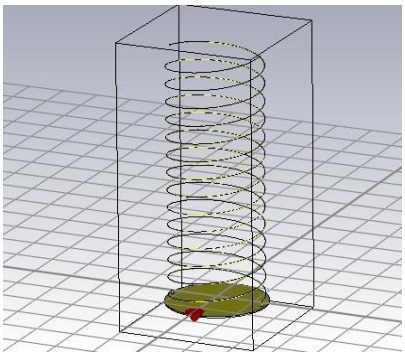
Frequency = 2.8 GHz
Main lobe magnitude = 7.65 dBi
Main lobe direction = 72.0 deg.
Angular width (3 dB) = 9.4 deg.
Side lobe level = -1.0 dB

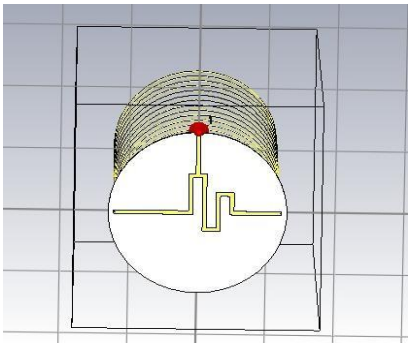
Impedance:



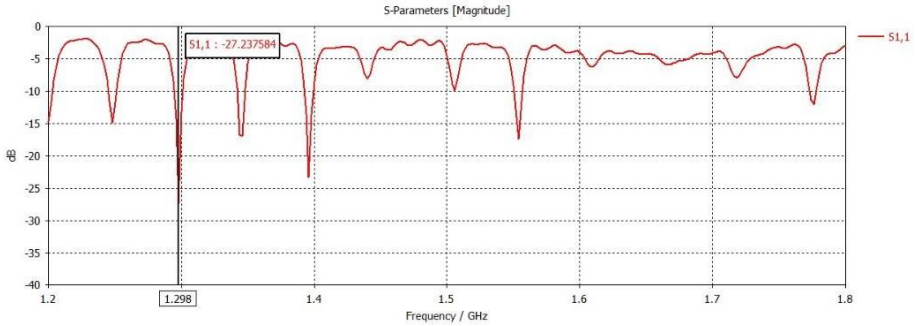
5. Dual Helical Antenna (with transmission line for 2λ):

3D Model:

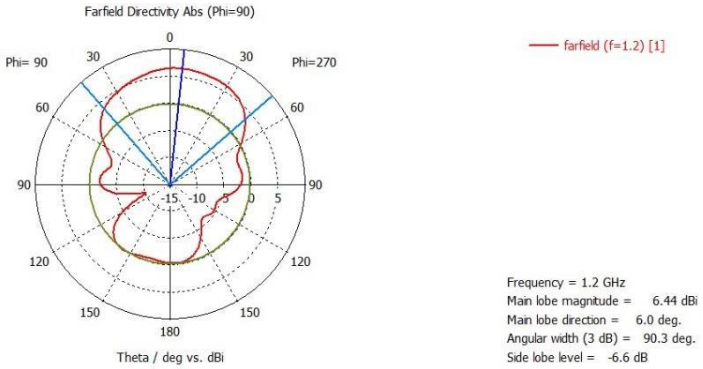




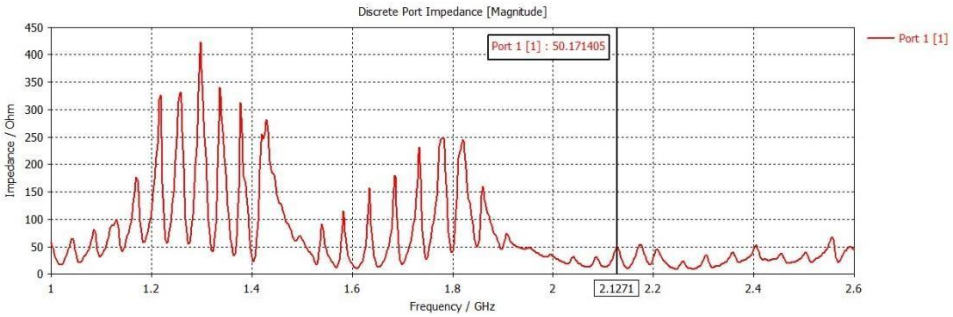
S-Parameter:



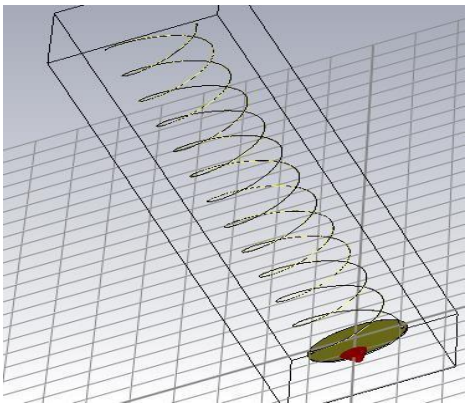
Farfield Directivity:



Impedance:

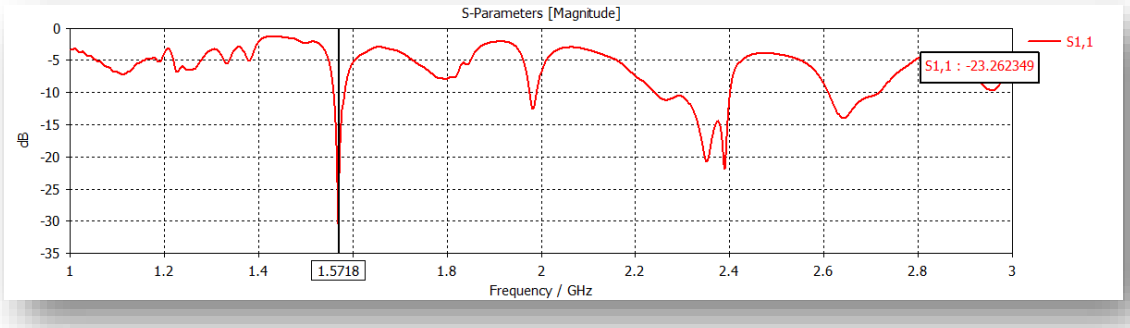


Modifying to r = 50mm and height to 343mm:

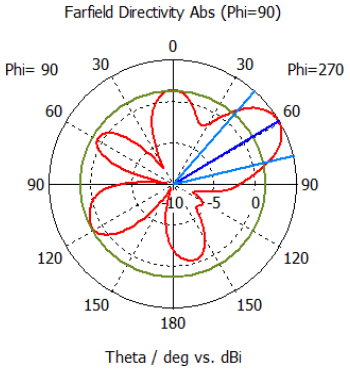


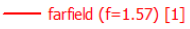
Parameters:
h = 343mm
N = 6
α = 15
s = 84.179mm
R = 50mm
R-wire = 0.5mm
R-gnd = 55mm

S-Parameter:



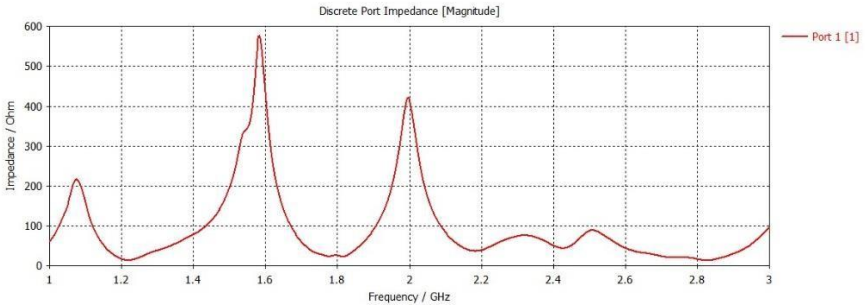
Farfield Directivity:





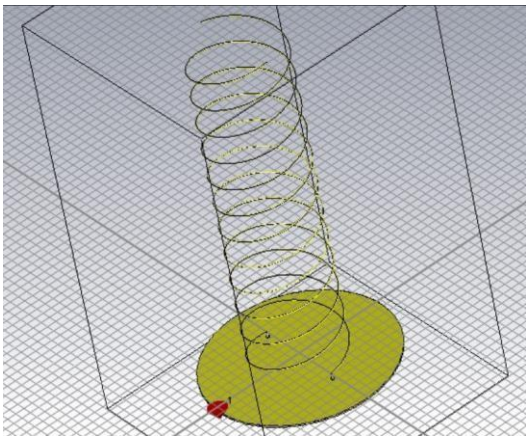
Frequency = 1.57 GHz
Main lobe magnitude = 4.84 dBi
Main lobe direction = 59.0 deg.
Angular width (3 dB) = 35.6 deg.
Side lobe level = -3.5 dB

Impedance:



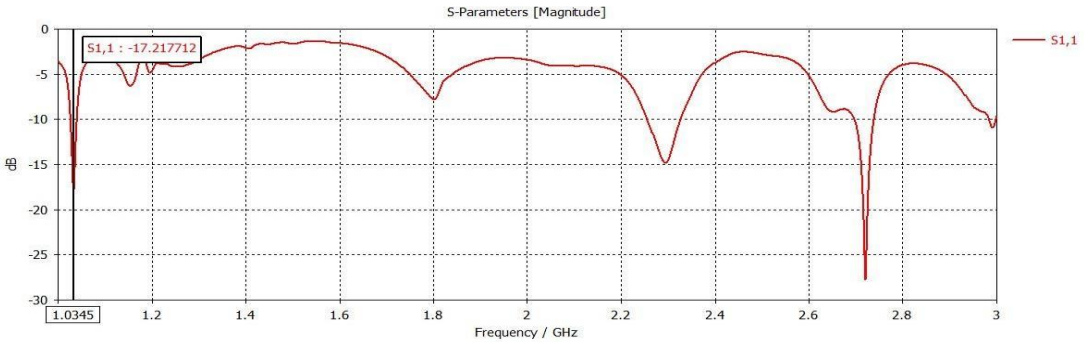
Modified GND radius to 100mm :

3D Model:

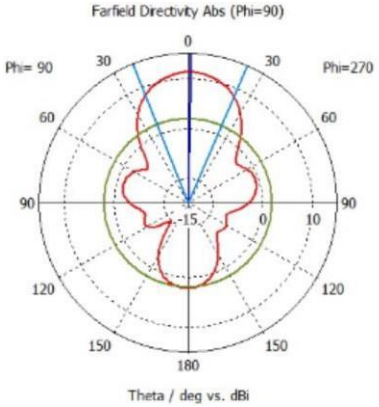


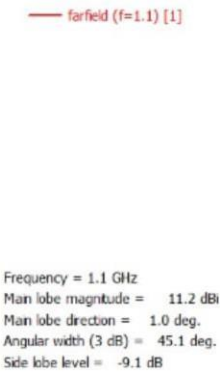
Parameters:
h = 343mm
N = 6
α = 15
s = 84.179mm
R = 50mm
R-wire = 0.5mm
R-gnd = 100mm

S-Parameter:

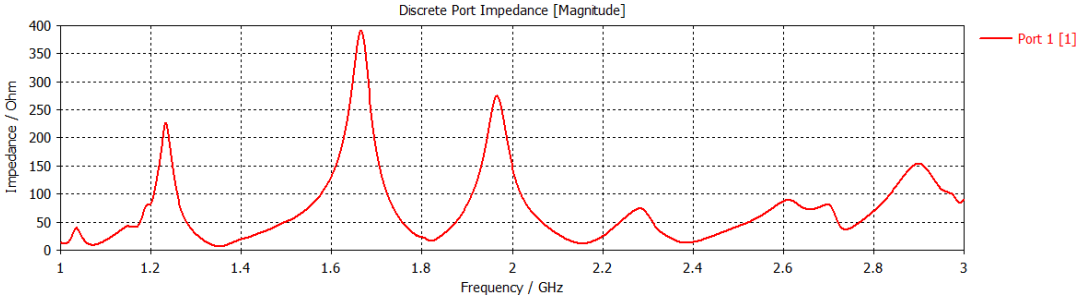


Farfield Directivity:





Impedance:



Conclusion:

In this project, the design and simulation of a Dual Helix Antenna were successfully carried out using CST Studio Suite (CST Microwave Studio). The antenna structure, consisting of two helically wound conductors, was analyzed with variations in helix dimensions such as radius, pitch angle, and number of turns.

From the design calculations, parameters were optimized for operation in axial mode, providing circularly polarized radiation along the antenna's axis a desirable characteristic for satellite and telemetry applications. The effective dielectric constant ($\varepsilon_{eff} = 3.33$) and guided wavelength ($\lambda_g = 104.35$ mm) were determined for accurate design of the transmission line and feed network.

Through simulation, the λ dual helical antenna achieved resonance at the desired operating frequency of 1.575 GHz (L1 band), confirming that the designed bifilar antenna meets the targeted performance requirements. The results indicate good impedance matching, wide bandwidth, and stable circular polarization characteristics.

Overall, the project successfully demonstrates the design principles and electromagnetic behavior of a dual-helix antenna at 1.57 GHz, validating its suitability for GPS L1 frequency band and similar high-frequency communication applications at the following parameters: h = 500mm, N = 6, $\alpha = 15$, s = 40.42mm, R = 30.33mm, R-wire = 0.5mm, R-gnd = 55mm

References

[1] C. A. Balanis, Antenna Theory: Analysis and Design, 4th ed. Hoboken, NJ: John Wiley & Sons, 2016.

[2] Dassault Syst`emes, “CST Studio Suite - Electromagnetic Field Simulation Software .